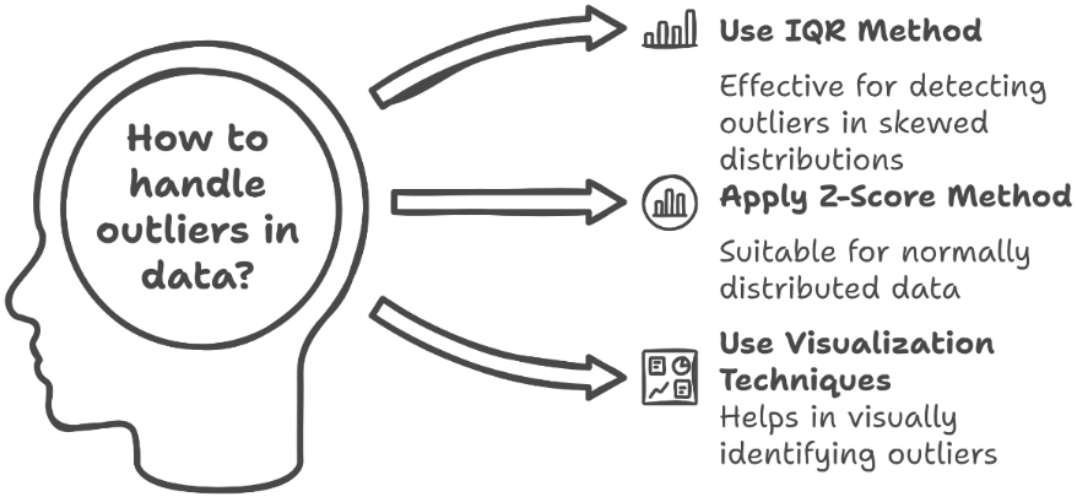
- **One-Hot Encoding:** Creates binary columns for each category:

```
encoded_data = pd.get_dummies(data, columns=['categorical_column'])
```

Addressing Outliers

Outliers can significantly impact statistical and machine learning models.



Detecting and handling outliers involves methods such as:

- **Using the Interquartile Range (IQR):**

```
Q1 = data['column_name'].quantile(0.25)
Q3 = data['column_name'].quantile(0.75)
IQR = Q3 - Q1

data = data[(data['column_name'] >= Q1 - 1.5 * IQR) & (data['column_name'] <= Q3 + 1.5 * IQR)]
```

- **Z-Score Analysis:**

```
from scipy.stats import zscore

data['z_score'] = zscore(data['column_name'])
data = data[data['z_score'].abs() < 3]
```

Data cleaning and preprocessing are crucial for transforming raw data into a structured format suitable for analysis and modeling. By addressing missing values, duplicates, incorrect data types, and scaling, you ensure that your dataset is clean and ready for meaningful analysis. With Python's Pandas